THE JUDGE PROTOCOL (TJP)

A Framework for Hardware-Enforced Human Oversight of Artificial Intelligence

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Preprint v1.0 | June 2026 | CC BY 4.0 / CERN OHL-S v2

arXiv categories: cs.CY (primary), cs.AI (secondary)

Zenodo DOI: 10.5281/zenodo.20401666 | github.com/judgeprotocol/protocol

Abstract

Artificial intelligence is now embedded in military targeting, financial markets, healthcare, and critical infrastructure — operating at speeds and scales no existing governance framework addresses.

This paper proposes the Judge Protocol: a global framework establishing a mandatory human authorization layer for all consequential AI actions.

The protocol's core principle — AI may analyze, recommend and simulate, but a human must authorize — is operationalized through a Decision Gate mechanism classifying AI actions into four tiers, from hard-stopped existential actions (Tier 1) to autonomously executed routine operations (Tier 4).

A hardware enforcement layer — the Hardware ACK (HACK) — implements the authorization requirement in chip firmware, below the software stack where governance cannot be bypassed.

The governance architecture proposes a two-layer structure: national AI authorities (NAIAs) responsible for compliance enforcement within member states, coordinated by an International Artificial Intelligence Agency (IAIA) modelled on the IAEA.

The protocol draws its architectural lineage from TCP/IP — no packet delivered without acknowledgement — and its institutional model from CERN and the Geneva Conventions.

The open standard is designed for adoption by any nation or AI system operator, owned by nobody, governed by all. This working document is offered as a rigorous starting point for the binding international governance framework that the current AI crisis demands.

Keywords: AI governance, human oversight, hardware enforcement, international governance, decision gate, autonomous systems

1. Introduction

The 2026 Gulf conflict has made undeniable what was already clear in principle: AI systems are embedded in active military operations without a binding global governance framework. Simultaneously, research from King's College London demonstrated that frontier AI models recommended escalation — including nuclear strikes — in 95% of simulated crisis scenarios, with no model choosing de-escalation [1].

In May 2026, Pope Leo XIV issued Magnifica Humanitas, calling explicitly for "robust legal frameworks" and declaring it "not permissible" to entrust lethal decisions to AI systems [2].

Existing frameworks — the EU AI Act, national regulations, sector-specific rules — address particular domains within particular jurisdictions. None establish a binding cross-domain, cross-border framework for human decision-making authority over consequential AI actions.

The governance gap is global in scope and widening faster than regulatory capacity can close it.

This paper proposes the Judge Protocol to fill that gap. Section 2 introduces the core principle and Decision Gate mechanism. Section 3 describes the hardware enforcement layer. Section 4 presents the governance structure. Section 5 outlines the implementation pathway. Section 6 positions the protocol relative to existing work. Section 7 identifies open questions and calls for collaboration.

2. The Core Principle and Decision Gate

The Judge Protocol is built on one principle: AI may analyze, recommend and simulate — but a human must authorize. This principle is operationalized through the Decision Gate — a mandatory human authorization checkpoint modelled on the Change Request processes already standard in enterprise IT and critical infrastructure worldwide.

The architectural model is TCP/IP: no packet is considered delivered without an explicit acknowledgement. In the Judge Protocol, no consequential AI action is considered authorized without an explicit human acknowledgement — an ACK token, cryptographically signed and verified by the enforcement hardware before compute capacity is released for execution. Figure 1 illustrates the direct mapping between the TCP/IP stack and the Judge Protocol stack.

Figure 1: The Judge Protocol stack mapped against TCP/IP. The ACK principle operates at Layer 4 in both architectures. TCP/IP delivers packets; the Judge Protocol authorizes AI actions. Same architectural principle — explicit acknowledgement before execution.

Every consequential AI action is classified into one of four tiers, shown in Table 1:

Tier	Classification	Examples	Authorization
Tier 1	Hard Stop — Existential	Nuclear, biological, chemical weapons; mass casualty actions	HARD STOP — no override. AI cannot recommend, simulate or execute.
Tier 2	High — irreversible, large-scale	Military strike authorization; critical infrastructure shutdown	Multi-party human authorization — minimum two independent authorities
Tier 3	Significant — material impact, partially reversible	Large financial transactions; clinical treatment decisions; targeting support	Single designated human authority must authorize
Tier 4	Routine — standard, reversible, low impact	Automated reporting; scheduling; routine analysis	AI executes autonomously — logged and auditable

Table 1: Decision Gate classification tiers.

An Emergency Change Process provides a time-critical authorization pathway for genuine operational emergencies while preserving human authority. Speed of authorization is an engineering problem; it is not a justification for removing the human authorization requirement.

Every Decision Gate event generates an immutable audit log entry containing the AI system identifier, action classification, impact assessment, human authorizer identity, decision timestamp, rationale, and outcome. Audit logs are retained for a minimum of ten years and are accessible to the IAIA governing body on request. Accountability is not aspirational — it is architectural.

3. Hardware Enforcement — The HACK

Governance implemented only in software can be patched, bypassed or overridden. The most robust implementation of the authorization principle is one realized below the software stack, in silicon.

The Hardware ACK (HACK) is proposed as a dedicated module co-packaged with AI accelerator chips. Built on the CV32E40S RISC-V core with an OpenTitan Chain of Evidence (COE), the HACK module performs four functions:

- Logs every Decision Gate event to tamper-proof on-chip memory that no software process can alter;
- Hard-stops Tier 1 actions in hardware logic — no software or firmware update can override;

- Holds a gate on compute capacity for Tier 2 and Tier 3 actions pending receipt of a verified, cryptographically signed human ACK token;
- Passes Tier 4 actions autonomously while logging to the immutable audit trail.

Figure 2: Decision Gate enforcement flow through the HACK chip. A single AI compute instruction traverses all four tier paths. Gate, crypto, audit log and Tier 1 hard stop are all silicon — no step in this chain can be shortcut by software.

The adoption pathway follows the Trusted Platform Module (TPM) precedent: open standard definition by a founding industry consortium, ratification as an ISO/IEC standard (analogous to TPM's ISO/IEC 11889), and progressive adoption by hardware manufacturers. The HACK specification is proposed for development under the CERN Open Hardware Licence (CERN OHL-S v2), with CERN as technical authority — the same institutional role CERN played in the development of the World Wide Web.

The enforcement model requires that authorization signals travel faster than the AI actions they govern. A photonic communication layer — proposed to run on the GÉANT research network infrastructure — provides the physical speed and tamper-resistance required for real-time global oversight. Signal interception requires physical disruption of the optical path, which is detectable by design.

4. Governance Structure

The Judge Protocol proposes a three-layer governance architecture designed to be enforceable without being centralized, and universal without requiring any state to surrender sovereignty. Figure 3 shows the full three-layer architecture from silicon to global governance.

Figure 3: End-to-end three-layer architecture. Layer 1 — silicon enforcement (HACK chip). Layer 2 — national authority (NAIA, Decision Gate Platform, human authorizer). Layer 3 — global governance (IAIA). Data flows upward only — never laterally between nations.

4.1 National AI Authority (NAIA)

Each member state establishes a National AI Authority (NAIA) headed by a ministerial-level Representative Lead. NAIAs maintain the national AI systems registry, audit Decision Gate compliance, manage AI incidents, enforce private sector obligations, and report compliance metadata to the global layer. National data remains under national jurisdiction at all times — sovereignty by design.

4.2 International Artificial Intelligence Agency (IAIA)

The International Artificial Intelligence Agency (IAIA) serves as a UN specialized agency modeled on the IAEA. Its General Assembly grants equal voting rights to all member states with no permanent-member veto. The IAIA operates the Global AI Registry — the central registry of all declared AI systems worldwide — and the Global Emergency Dashboard, a real-time three-level alert system (Normal / Elevated / Crisis) publicly visible at summary level. No single actor can override the alert level; it is set algorithmically.

Figure 4 illustrates the global NAIA network and the upward-only data flow architecture.

Figure 4: Global IAIA/NAIA network. Compliance metadata flows upward from national authorities to the IAIA — never laterally between nations. Sovereignty by design.

4.3 Design Principles

Three design principles address the central political objections to global AI governance, and are built into the architecture rather than stated as aspirations:

- **Sovereignty by design:** national data remains under national jurisdiction; the global layer aggregates only compliance metadata, never raw content.
- **Human authority always:** no operational urgency exception exists — the Emergency Change Process is the time-critical pathway, not an override.
- **Evolution by design:** mandatory review cycles and a defined governance process ensure the protocol evolves as AI capability evolves.

The IAIA coordinates with — but does not replace — existing bodies: the IAEA on nuclear-related AI, WHO on healthcare AI, BIS and FSB on financial AI, and ITU on technical standards. The Judge Protocol fills the governance vacuum between existing frameworks; it does not duplicate them.

Funding follows the CERN model: member state contributions proportional to GDP, complemented by an AI chip micro-levy at point of manufacture and voluntary contributions from major AI platform operators. No single source exceeds 15% of total IAIA budget. Industry contributions carry no voting rights or policy influence.

5. Implementation Pathway

The Judge Protocol is implemented in four phases, each proving the model at the previous scale before extending. The governing principle is: prove before mandate. The sandbox generates the evidence base that justifies mandatory adoption.

- **Phase 1 — Sandbox (0–12 months):** Decision Gate Platform deployed within a founding national NAIA, operating against a single domain. Live audit trail, ACK latency data, and human override rate data published as evidence base for v1.0.
- **Phase 2 — Reference Implementation (12–24 months):** Software stack packaged as The Judge Protocol Image — open source, deployable by any NAIA on commodity hardware. IAIA interim secretariat established.
- **Phase 3 — Hardened Deployment (24–48 months):** TMO PCIe enforcement card deployed. Air-gapped appliance specification published under CERN OHL. Full domain rule set coverage across all twelve domains.

- **Phase 4 — Silicon Enforcement (48–72 months):** HACK core embedded in AI accelerator silicon. Government procurement mandates requiring HACK compliance adopted by member states. Universal silicon-level AI governance enforcement.

6. Related Work

The EU AI Act [3] establishes a risk-based regulatory framework within EU jurisdiction but does not propose binding cross-border human authorization requirements or hardware enforcement mechanisms.

The Rome Call for AI Ethics [4], co-signed by Microsoft, IBM, the UN Food and Agriculture Organization and others, establishes ethical principles including human oversight but is a voluntary declaration, not an enforcement mechanism.

The Bletchley Declaration [5] represents multilateral acknowledgement of frontier AI risk by 28 governments, but does not establish binding enforcement mechanisms or a permanent governance body.

RAND and CNAS work on Hardware-Enabled Mechanisms (HEMs) [6, 7] addresses hardware governance in the context of AI chip export controls — a distinct but complementary problem. The Judge Protocol addresses AI action governance: not where chips go, but what AI systems do with them.

Ansari (2026) [8] proposes a taxonomy of 20 hardware-level governance mechanisms assessed for technical feasibility. That paper taxonomizes possible mechanisms; the Judge Protocol proposes a complete governance framework with a specific implementation pathway, institutional architecture, and open standard.

The Judge Protocol's novel contribution is the combination of: (1) a universal human authorization layer applicable across all domains and jurisdictions; (2) hardware enforcement at silicon level below the software stack; and (3) a two-layer governance architecture with a global body and national implementation authorities. No existing framework combines all three elements.

7. Open Questions and Call for Collaboration

Several areas require expert collaboration before v1.0 ratification:

- **Military and Defense (Appendix F3):** The wartime exception is architecturally the most sensitive element. It must be developed in collaboration with the ICRC, UN Group of Governmental Experts on Lethal Autonomous Weapons Systems, and member state defense establishments before v1.0 ratification. Unconditional hard stops may limit military adoption; carelessly defined exceptions undermine the entire framework.
- **Hardware specification:** HLD Sections 4–8 covering detailed HACK hardware specifications are pending CERN Openlab engagement.
- **Founding sandbox partner:** A founding national NAIA partner is sought to deploy the Phase 1 sandbox and generate the evidence base for v1.0.
- **Domain rule sets:** Companion working papers for Financial Markets, Healthcare, Judicial, and Transport domains are in development.
- **Redirect Gate:** Beyond the Decision Gate's authorization function, the protocol defines a Redirect Gate — a generic escalation mechanism for actions that must be stopped and routed to a competent external authority rather than authorized internally, covering three confirmed use cases: Human Life Impact, Cyber Misuse, and AI Misuse for Private Gains. Use-case rule sets are still in development and require input from the relevant sector regulators and data protection specialists before ratification.
- **Physical Domain (robots and autonomous systems):** The full protocol extends the governance model to AI systems with physical agency — industrial, service, defense, and private consumer robots — including an unconditional hard line on autonomous weapons systems. This domain requires dedicated collaboration with military legal experts, the ICRC, and the UN Group of Governmental Experts on Lethal Autonomous Weapons Systems before v1.0 ratification.

 All documents, including the full protocol (v0.95) and Technical Architecture HLD (v0.67), are available at github.com/judgeprotocol/protocol and archived at Zenodo (DOI: 10.5281/zenodo.20401666). Correspondence: judgeprotocol@pm.me

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